

Feasibility of Real Time 3D Object Detection for Accessibility Use Cases

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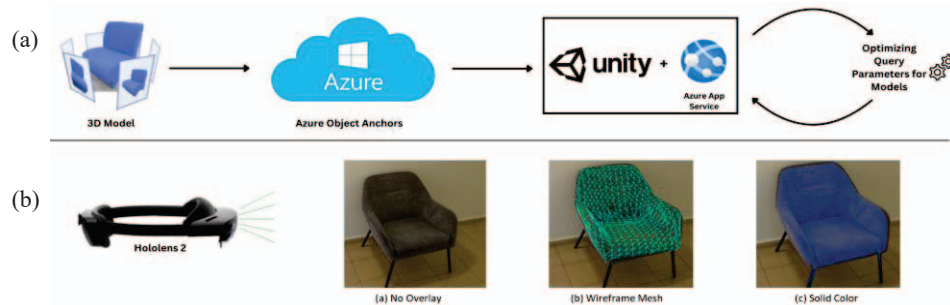


Figure 1: (a) Application process (b) Different styles of digital overlay

ABSTRACT

Augmented Reality (AR) holds promise for improving visibility and accessibility of real-world objects. AR systems, however, often rely on 2D overlays or marker-based methods, which face limitations such as camera's poor performance in low-light. We present a new approach to improve visual accessibility in AR. We developed a Unity-based application for the HoloLens 2, integrating Azure Object Anchors (AOA) for real-time 3D object detection with depth sensors. Using pre-scanned 3D models, the application detects and then projects virtual overlays on objects to increase their visibility while also logging eye gaze and spatial movement to enable evaluation of user interactions.

Keywords: Augmented Reality, Accessibility, Visual Impairments

Index Terms: Human centred computing, Perception, Augmented Reality

1 INTRODUCTION

A long-term promise of augmented reality (AR) is to increase accessibility of real-world environment by detecting objects and enhancing their visibility by overlaying digital information onto the real world. This could assist communities, who have deteriorated vision including people with low vision and older adults. Indeed, vision enhancement with AR has been demonstrated as an effective tool for daily tasks such as shopping, reading text, and mobility for people with low vision [1]. However, previous approaches have several limitations.

Some approaches used marker-based methods (e.g. QR codes) to detect objects, rather than real-time object detection. For example,

a recent study used QR codes to mark obstacles in AR on the HoloLens for low vision users, but this method had limited efficacy on user's mobility and required pre-marking QR codes on objects [2]. Marker-based approaches, rely on RGB cameras and require placing markers to function [3]. These approaches are often impractical for real-world applications where marker placement cannot be guaranteed. Approaches that developed object detection applications on platforms such as the HoloLens 2, typically rely on RGB cameras to detect objects [4-5]. These methods are dependent on lighting conditions, making them less reliable in dim environments.

Another limitation of previous research lies in the design of 2D digital overlays, such as text annotations or 2D bounding boxes. These designs add virtual elements to the scene, which needs to be perceived and processed in addition to the environment, instead of modifying object visibility directly. Naturalistic design of object enhancement in AR may provide a better experience for users, by avoiding extra virtual annotations, which could overwhelm users with limited visual capacities [2].

To overcome these limitations, we present a new approach to improve visual accessibility that could be examined in future user studies. We developed an application that detects real-world objects in real time using depth sensors and enhances objects' visibility in 3D. The application also captures user interactions and can provide a test bed for examining the utility of 3D object enhancements with depth sensors for different scenarios for low vision or older adult users. A key scenario we had in mind is improving mobility for low vision users by detecting and enhancing obstacles on their walking route. Additional scenarios are highlighting items in the kitchen while cooking [6], helping the search of an object, and home organization. We present how the application works, how to optimize it for low vision user scenarios, and considerations for future user studies.

2 METHODS

We developed an application for the HoloLens 2 to detect objects using the depth sensor and 3D models of objects, augmenting them with virtual mesh overlays. The application was built using Unity to interface with the Azure Object Anchors (AOA) SDK, managing object detection, rendering of mesh (see Fig. 1A), and capturing user behaviour. AOA enables recognition and tracking of objects by training a proprietary deep learning model to detect pre-

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processed 3D models in the environment. During runtime, AOA processes spatial data captured by the HoloLens 2's depth sensor to locate objects and return their position, orientation, and dimensions. This information is used to ensure accurate alignment of overlays with the detected objects (see Fig. 2). The specific algorithms and techniques AOA employ for detection and alignment are proprietary and are not publicly disclosed.

To use AOA, detailed 3D models of potential objects are required. The process begins by capturing images of objects from multiple angles to generate high-fidelity 3D models using depth sensors or other compatible devices. These models are manually refined and uploaded to AOA's cloud training and conversion service, which processes them into a HoloLens-compatible format for runtime detection. To maximize detection accuracy, our application utilized configurable AOA parameters. The search area was configured to align with the user's field of view, ensuring correct object projection in the immediate environment. We modified dimensional thresholds (width, height, and depth) to focus on specific furniture while reducing false positives, and adjusted surface coverage thresholds to detect only fully visible objects. Additionally, we employed AOA's low-latency object tracking mode for stability during user movement.

Different user scenarios are likely to require different visual highlights across objects and user tasks. Thus, our application was also designed to compare the efficacy of different virtual overlays (see Fig. 1B). For example, wireframe meshes may be used for lightweight and unobtrusive visualizations, solid color models can provide high-visibility overlays and 3D realistic models of the object may improve their natural representation (e.g., objects that require manual interactions may require light mesh overlay versus obstacles to avoid, which may require high visibility overlay).

Our goal is to test our application with user studies in real scenarios of visual struggle. Thus, our application was designed to log critical user data during runtime: eye gaze data, spatial movement, and orientation within the environment. In addition, the application tracks key parameters related to the digital content, including the position of overlays, inference times for object detection, scale adjustments applied to match the detected objects, and the coverage ratio, which represents the proportion of the object's surface seen by the HoloLens' depth sensor.

3 LIMITATIONS AND FUTURE WORK

During the development and testing of our application, we encountered several limitations that should be investigated in future studies. One key challenge lies in its reliance on depth-sensing technology for detection, without integrating RGB data. This dependency can lead to difficulties in distinguishing objects with similar shapes or planar surfaces, particularly when multiple objects share overlapping geometric features. Additionally, size

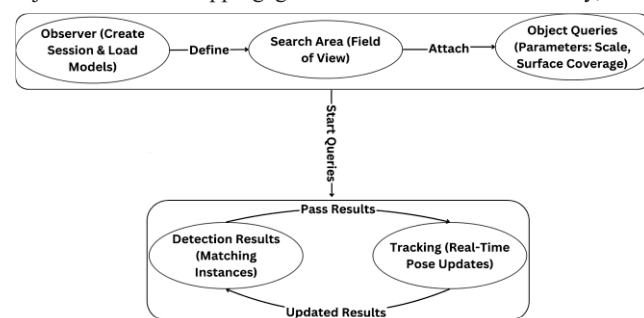


Figure 2: Process of Azure Object Anchors

limitations were observed during testing, objects with dimensions under 70 cm were harder to detect reliably due to the resolution constraints of the depth sensor. Future studies should consider

quantitative evaluation of these errors. Moreover, while the system successfully detected the first object in under one second, performance declined with each additional detection. After identifying five to six objects, no new detections occurred, this limitation, likely linked to current processing capacity, may be addressed in future iterations of AOA. Further experimentation revealed that while the HoloLens 2 encountered challenges under extremely low-light conditions, it remains effective in dim scenarios, thereby preserving its potential to offer enhanced capabilities beyond mere image-based detection methods. Finally, the current process requires pre-scanning and training of 3D objects ahead of time; while it enables high-fidelity object representation and precise alignment, it is a bottleneck for detection of new objects, although future advancements may eliminate this process.

During informal testing, we found that our various overlay styles successfully enhanced object visibility while preserving their natural appearance. The 3D overlays made objects distinctly noticeable without obstructing their original shape, possibly providing more intuitive understanding compared to traditional 2D approaches, that could be tested in future formal experiments. Additionally, our method addresses inherent challenges of 2D recognition and highlighting, such as unreliable depth estimation and reduced performance in low-light conditions, which are critical for enhancing accessibility in low-vision use cases. In sum, we present the potential and limitations of advanced real time 3D detection and alignment technologies with depth sensors through a practical AR application that can be tested in a variety of accessibility user scenarios.

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